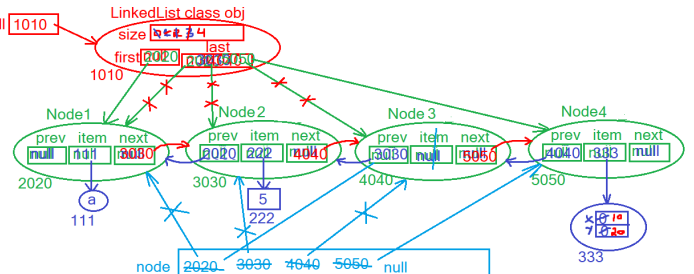
```
LinkedList<Object> ll = new LinkedList();
```

```
ll.add("a");
```

1) new node object is created with pass element
2) this node is linked to previous node and previous node is linked to this node
3) first node should store in first variable in LinkedList object
4) every node must store in last variable in LinkedList object because every node is a last node




```
Node n1  = new Node(e, last, null);
```

```
class Linked {
```

```
    ll.add(5);
    ll.add(null);
    ll.add(new Example(5, 6));
```

```
class Node {
    private Object item;
    private Node prev;
    private Node next;
    Node(Object item, Node prev, Node next) {
        this.item = item;
        this.prev  = prev;
        this.next  = next;
    }
}
```

```
}
```

```
clas LinkedList {
    ---
    ----
    ----
    void add(E e) {
        Node newNode = new Node(e, last, null);

        if(first == null)
            first = newNode;
        else
            last.next = newNode;
        last = newNode;
        size++;
    }
}
```

```
int indexOf(Object obj) {
    Node node = first;

    for(int i=0; i<size; i++) {
        if(obj.equals(node.item))
            return i;
        node =  node.next ;
    }
    retrun -1;
}
```

```
Object get(int index) {
    if(-----)
      throw -------

    if(index < size/2) {
        Node node = first;
        for (-----) {
          -----
          node = node.next;
        }
    else {
        Node node = last;
        for(-------) {
          ----------
          node = node.prev;
        }
    }
```

```
Object get(int index)  {
    if(index <0 || index>=size)
      throw new IOBE(index);
    Node node = first;
    for(int i = 0; i<size; i++){
        node = node.next;
        if(i == index)
          return node.item;
    }
}
```

Linear Search
Wrong design